Section 1.2

11.

Stem	Leaf
6L	4 3 0
6H	7 6 9 6 8 9
7L	4 2 0 1 4 2 0 2
7H	
8L	0 1 1 2 1 1 4 1 0 3 4 2
8H	9 5 9 5 5 7 8
9L	3 0
9H	5 8

Stem: ten's digit with low or high leaves.

leaf: one's digit.

Feature: It is negatively skewed. The data concentrates around 7L to 8H mainly.

The scores locating in the low 80s are most and locating in the high 70s are least. (nobody).

14. a. Stem: integer digits Leaf: decimal digits

Stem	Leaf
2	3 2
3	4 7 3 9 5 4 6 7 2 8
4	6 0 8 3 8 8 1 5 9
5	1 1 6 8 0 4 0 4 5 0 6 1 6 1 6 5 9 7 0
6	7 9 4 2 6 4 5 3 2 0 9 6 1 0 7 2 7 2 4 6 9 8 9 2 0 3 0
7	1 0 5 5 6 3 5 5 6 2 2 4 3 0 0 5 8 0
8	0 8 3 2 4 4 4 3 2
9	2 6 8 3 2 0 5 3 7 5 3 7 6 3 6 3 8 1
10	5 4 8 3 4 2 4 2 5 8 4 6
11	5 2 9 3 9 9 3
12	3 7
13	8
14	3 6
15	0 3 5 0
18	9

b. The flow rate concentrated around 6.0 – 7.0 L/min, Thus take the median 6.5 L/min as a typical flow rate.

c. The data is highly concentrated around 5.0 – 10.0 L/min and spread out a little due to extreme value 2.2, 2.3 and 18.9.

d. It is not symmetric but positively skewed. Since the right tail is stretched out compared with left tail.

e. 18.9 L/min is an outlier. Since the second highest data is 15.5 L/min. 18.9 L/min is far from the main body.

20.

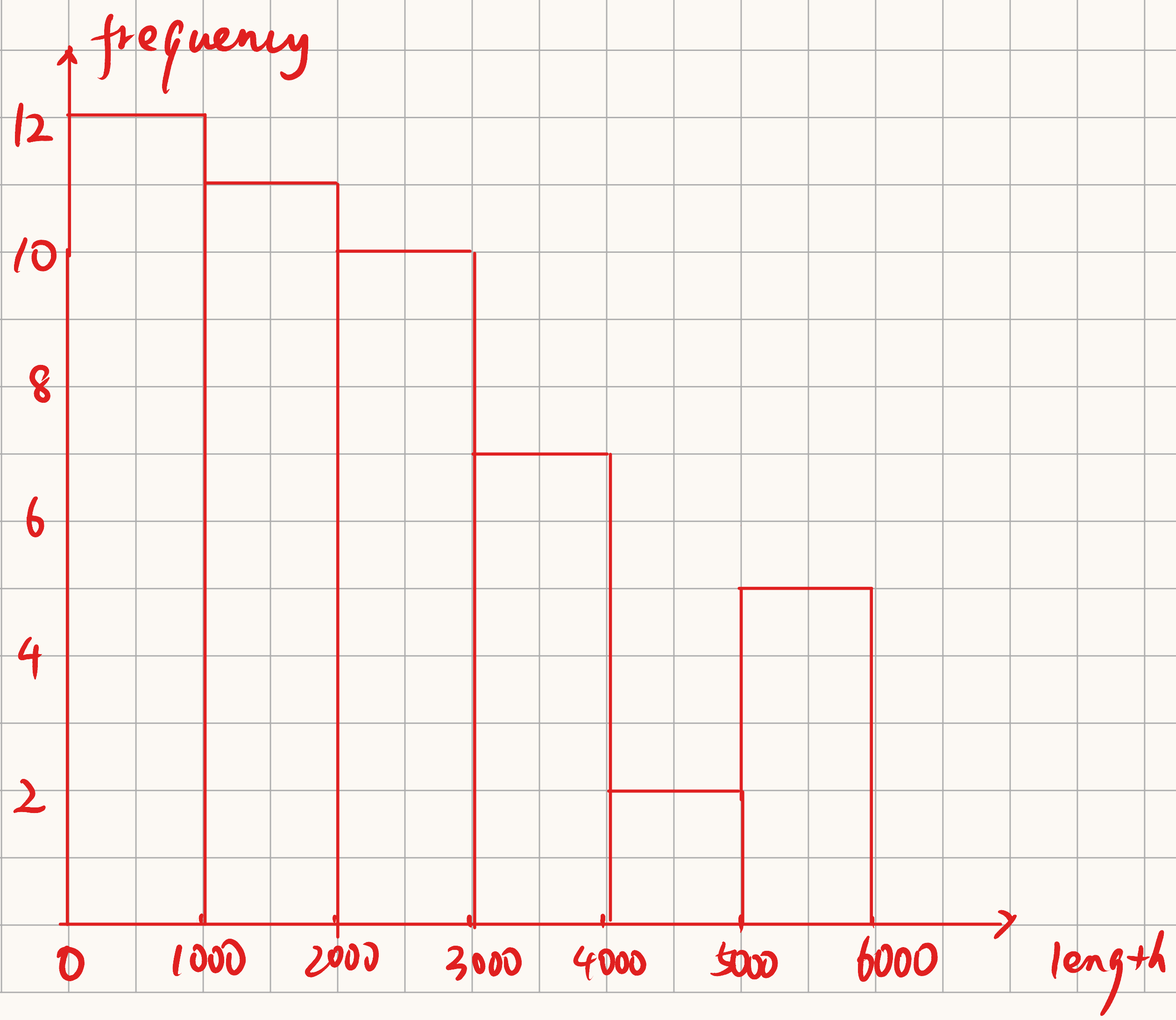
a. Stem: thousands digit

Leaf: hundreds digit

Stem	Leaf
0	3 3 9 5 5 9 4 5 1 5 2 3
1	2 2 0 0 3 2 1 8 6 8 4
2	1 4 1 2 3 4 4 7 7 1
3	0 3 3 3 8 1 1
4	3 7
5	3 7 2 8 7

Features: It is positively skewed. The data concentrates on 0 thousands to 2 thousands mainly.

b.



proportion $(l<2000)$: $f_1 = \frac{12+11}{47} \approx 0.489$

$(2000 \le l < 4000)$: $f_2 = \frac{10+7}{47} \approx 0.362$

The shape of the histogram is unimodel and positively skewed.

Section 1.3

34.

a. $\bar{x}_u = \frac{1}{11}\sum_{i=1}^{11} x_{ui} = \frac{1}{11}\times(6.0+5.0+\cdots+23.0) = 21.55$

$\bar{x}_F = \frac{1}{15}\sum_{i=1}^{15} x_{Fi} = \frac{1}{15}\times(4.0+14.0+\cdots+0.3) = 8.56$

And $\bar{x}_u > \bar{x}_F$

b. After ordering,

U: 4.0 5.0 5.0 6.0 11.0 17.0 18.0 23.0 33.0 35.0 80.0

F: 0.3 2.0 3.0 4.0 4.0 5.0 8.0 8.9 9.0 9.0 9.2 11.0
14.0 20.0 21.0

Thus $\tilde{x}_u = 17.00$ $\tilde{x}_F = 8.90$

And $\tilde{x}_u > \tilde{x}_F$

The extreme value 80.0 affects the mean, making mean is so different from the median.

c. For U: $\lambda_1 = \frac{1}{11}\times 100\% \approx 9.09\%$

For F: $\lambda_2 = \frac{1}{15}\times 100\% \approx 6.67\%$.

And $\bar{x}_{u\,tr(9.09)} = \frac{1}{9}\sum_{i=2}^{10} x_{ui} = \frac{1}{9}\times(5.0+5.0+\cdots+35.0) = 17.00$

$\bar{x}_{F\,tr(6.67)} = \frac{1}{13}\sum_{i=2}^{14} x_{Fi} = \frac{1}{13}\times(2.0+3.0+\cdots+20.0) = 8.24$

$\bar{x}_u > \bar{x}_{u\,tr(9.09)} = \tilde{x}_u$

$\tilde{x}_F > \bar{x}_F > \bar{x}_{F\,tr(6.67)}$

40.

$$\tilde{x} = \frac{91+93}{2} = 92.00$$

$$\bar{x}_{tr(25)} = \frac{1}{26}\cdot\sum_{i=13}^{38} x_i = \frac{1}{26}\times(65+67+\cdots+141) = 95.38$$

$$\bar{x}_{tr(10)} = \frac{1}{40}\sum_{i=6}^{45} x_i = \frac{1}{40}\times(36+38+\cdots+248) = 102.22$$

$$\bar{x} = \frac{1}{50}\sum_{i=1}^{50} x_i = \frac{1}{50}(11+14+\cdots+513) = 119.26$$

$$\bar{x} > \bar{x}_{tr(10)} > \bar{x}_{tr(25)} > \tilde{x}$$

Section 1.4

44. a. Order these values:

23.5 26.3 28.0 28.2 29.4 29.5 30.6 31.6 33.9 49.3

The sample range is $49.3 - 23.5 = 25.8$

b. $\bar{x} = \frac{1}{10}\sum_{i=1}^{10} x_i = \frac{1}{10}(23.5 + 26.3 + \cdots + 49.3) = 31.03$

$s^2 = \frac{1}{9} \cdot \sum_{i=1}^{10}(x_i - \bar{x})^2 = \frac{1}{9} \times [(23.5-31.03)^2 + (26.3-31.03)^2 + \cdots + (49.3-31.03)^2]$

$= 49.31$

c. $s = \sqrt{s^2} = 7.02$

d. $S_{xx} = \sum_{i=1}^{10} x_i^2 - \frac{(\sum_{i=1}^{10} x_i)^2}{10} = 443.80$

$s^2 = \frac{1}{9} S_{xx} = 49.31$.

56. Order the data:

15.30 16.20 16.35 17.15 17.48 17.73 17.75 17.85 18.00
18.68 18.82 18.85 19.03 19.07 19.08 19.17 19.20 19.20
19.33 19.37 19.43 19.48 19.50 19.58 19.60 19.62 19.90
19.97 20.00 20.05 21.22 22.25 22.75 23.25 23.78

The smallest value $= 15.30$ the largest value $= 23.78$

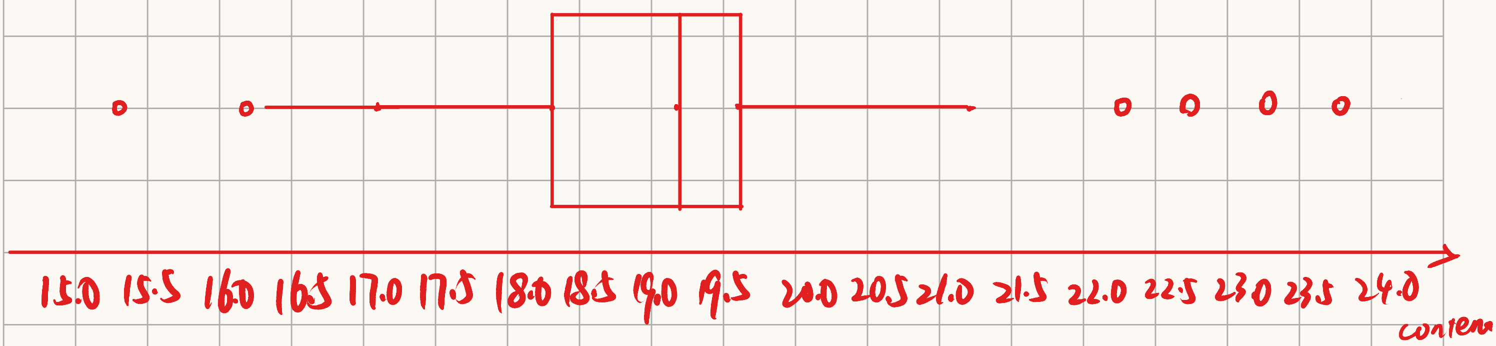
$\tilde{x} = 19.20$ the lower fourth $= 18.34$

the upper fourth $= 19.76$

And $f_s = 1.42$ And $1.5 f_s = 2.13$

And the distances of 15.30、16.20 to 18.34 is larger than $1.5 f_s$. The distances of 22.25, 22.75, 23.25, 23.78 to 19.61 is larger than $1.5 f_s$.

Thus 15.30、16.20、22.25、22.75、23.25、23.78 are outliers.



the sample mean $\bar{x} = \frac{1}{35}\sum_{i=1}^{35} x_i = \frac{1}{35} \times (15.30 + \cdots + 23.78) = 19.26$

the sample variance $s^2 = \frac{1}{34} \cdot \left(\sum_{i=1}^{35} x_i^2 - \frac{(\sum_{i=1}^{35} x_i)^2}{35}\right) = 3.36$

Thus. the data concentrates around 18.0-19.6 mainly. And it is positively skewed. The percentage - outliers is $\frac{6}{35} \times 100\% = 17.14\%$ thus the variation is affected mainly by extreme data.